ANANYA GUPTA

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PROFILE

AI/ML Engineer specializing in Generative AI and multimodal systems, with a strong foundation in research and production-grade AI. **B.Tech in AI & Data Science (CGPA 9.2), author of 2 international research publications and 1 registered design patent**. Experienced in building **LLM-powered systems including RAG chatbots, document intelligence pipelines, and government-facing AI solutions**, with proven ownership from **POC to production integration**. Strong expertise in AWS Bedrock, vision–language models, vector search, and prompt orchestration, bridging rigorous experimentation with real-world deployment.

EDUCATION

Vivekananda Institute of Professional Studies – Technical Campus, New Delhi, India
Bachelor of Technology in Artificial Intelligence and Data Science | CGPA 9.2/10.0 **2021 – 2025**

Kulachi Hansraj Model School, New Delhi, India
Senior Secondary Examination(CBSE) | 86% **2007 – 2021**

PUBLICATIONS & PATENTS

Accepted Papers:

- **Emotion Aware Video Summarization System** — Design Patent No. 451236-001 | **LINK** **Sep'25**
- **Unveiling Offensive Content Detection: An Experimental Study using Deep Learning and Machine Learning Techniques**
  ICDAM2024, Springer- LNNS Series **| LINK** **June'24**
- **Machine Learning and Natural Language Processing Strategies for Fake News Detection: An Empirical Study**
  AIAMMS2023, Springer - CRC Press / Taylor & Francis **| LINK** **Nov'23**

EXPERIENCE

Biz Digital IT | Gurugram, India **Jan'25 - Present**
AI Software Engineer| Hybrid

- Designed and implemented a **government-facing RAG chatbot for MIDC** using **AWS Bedrock Knowledge Base and Amazon Nova Pro**, owning the system **from POC to production integration**.
- Implemented **semantic retrieval, multi-turn follow-up handling, and hallucination control**, ensuring accurate, scope-restricted responses aligned with official MIDC data.
- Designed and deployed **two production Generative AI systems** for **document summarization and question answering** across PDFs, DOCX, images, and structured data using **GPT-4o (Azure OpenAI)**.
- Optimized **multi-document processing pipelines** using parallel execution, reducing end-to-end latency from **~40s to ~10s**.
- Developed **token-level inference analysis** (per page and per document) to enable **LLM cost monitoring and optimization**.
- Conducted **research-oriented experimentation** on complex **multi-page invoice understanding**, evaluating **vision–language models, OCR–LLM hybrids, and structured extraction pipelines**.
- Designed and implemented a **RAG framework on AWS** using **Amazon Nova Pro** for long-document summarization and **Amazon Titan embeddings** for semantic retrieval over large document collections.
- Currently developing a **cost-efficient invoice extraction pipeline** using **Amazon Nova Pro**, achieving **~93–95% accuracy**, with ongoing optimization and evaluation.

Docquity | Gurugram, India **July'24 – Sept'24**
R&D Intern | on-site

- Fine-tuned **Phi-1.5** on large-scale biomedical text (PubMed XML).
- Explored **prompting and inference behavior** across Claude, OpenAI, and AWS Bedrock models.
- Gained exposure to **LLM serving, scalable inference pipelines, and system-level architecture**.

SKILLS

- **Generative AI & LLMs:** RAG, Prompt Engineering, LLM Orchestration, GPT-4o, Claude 3, Amazon Nova Pro, Qwen2.5, Ollama, Llama CPP
- **ML / NLP: Transformers,** Vision–Language Models, Text Classification, Information Extraction, Topic Modeling
- **Programming:** Python
- **Frameworks & Libraries:** PyTorch, TensorFlow, Scikit-learn, Hugging Face
- **Vector & Data Stores:** FAISS, Vector Databases, SQL, MongoDB, SQLAlchemy
- **Cloud & Tools:** AWS (Bedrock), Azure OpenAI, Git, GitHub, Jira, Groq Cloud
- **Data & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, BeautifulSoup

PROJECTS

- **A Unified Hybrid Framework for Sentiment Analysis from Visual-Textual Modalities Using Nova Inference and Hierarchical Deep Learning** **Feb'25 – May'25**
  Tools: AWS Bedrock , PyTorch, Hugging Face (BERT, RoBERTa, DistilBERT), Python
  - Built a system that analyzed sentiment from raw images and text by generating structured metadata with **Amazon Nova Pro**. Fine-tuned **BERT, RoBERTa, and DistilBERT** (via **Hugging Face**) using **PyTorch** and **Torch**, achieving **97% accuracy**
  - Created vibrant visualizations (e.g., **radial heatmaps, tree maps**) with Python/PyTorch, producing publication-ready figures for a research paper.
- **Medical RAG Chatbot for Knowledge-Grounded Question Answering | Link** **Aug'24 – Sept'24**
  Tools: Sentence Transformers, FAISS, RAG, Groq LLM, Python, Streamlit
  - Designed a **Retrieval-Augmented Generation (RAG) system to answer domain-specific queries** using a **structured CSV-based** knowledge base.
  - Implemented semantic retrieval with Sentence Transformers and **FAISS** to ground LLM responses and **reduce hallucinations**.
  - Engineered modular preprocessing, **retrieval**, and **generation** components with transparent context inspection and **conversational memory**.
- **Multi-File RAG & LLM Chat System with Streamlit | LINK** **Sept'24 – Dec'24**
  Tools: Grok API , Hugging face, sentence transformers, Faiss, RAG, python, Llama 8B, Python, Streamlit
  - **Developed a Streamlit application** integrating **Retrieval-Augmented Generation (RAG)** to support multi-file uploads (PDF, DOCX, TXT, MP3, MP4) for context-based responses.
  - **Implemented a conversational interface** using **Llama 8B via the Groq API**, enabling users to engage in natural language queries with the model.
  - **Enhanced user experience** by allowing seamless interaction with uploaded documents, facilitating efficient information retrieval and summarization.

CERTIFICATES

- Data Bricks Certified Data Engineer Associate | **LINK** **Jan'26**
- AWS Certified AI Practitioner| **LINK** **July'25**
- Research Paper Presentation Certificate, ICDAM2024 **June'24**
  (5th International Conference on Data Analytics
  and Management (ICDAM-2024)) | **LINK**
- Research Paper Presentation Certificate, AIAMMS2023 **Nov'23**
  (1st International Conference on Artificial Intelligence, Advanced Materials, and
  Mechatronics Systems (AIAMMS-2023))| **LINK**
- NODEJS - IBM | **LINK** **Oct'23**